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College: DY Patil

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective

To understand Azure cloud concepts and build a complete data pipeline using Azure Storage Account and Azure Data Factory.

Assignment Tasks

Task 1: Explore Azure Portal

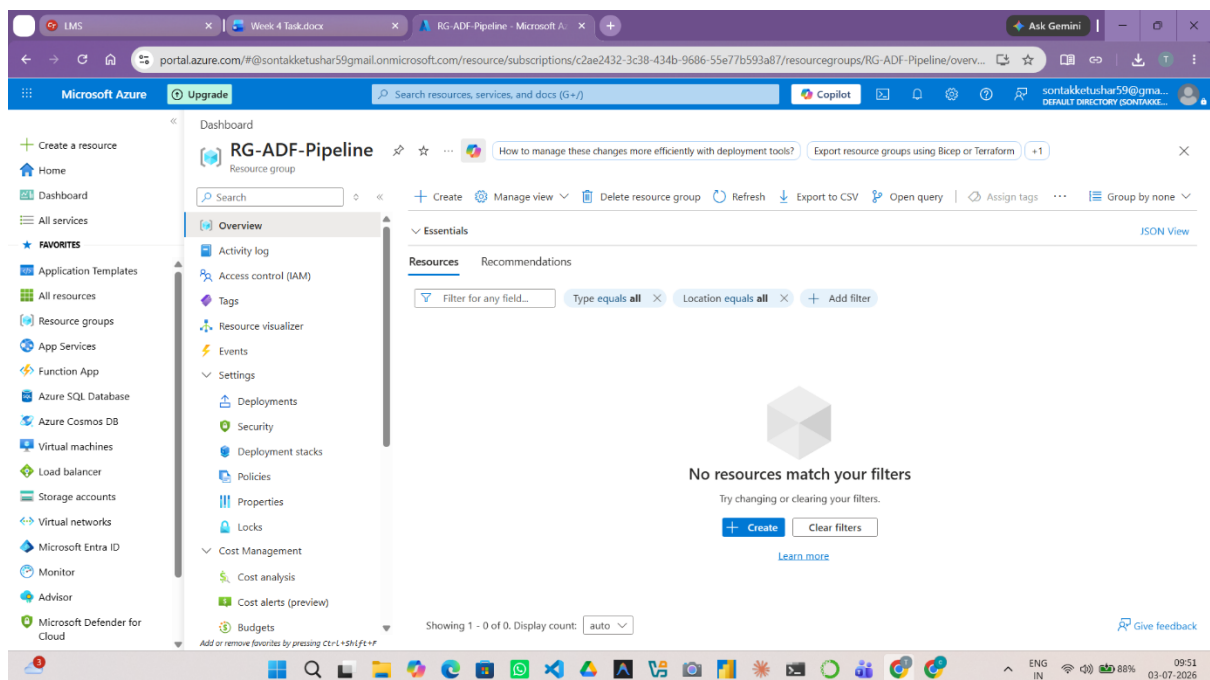
Do:

- Explored Azure Portal.
- Created a Resource Group.

Deliverable:

Description: Created a Resource Group to organize Azure resources.

Screenshot:



Task 2: Storage Setup

Do:

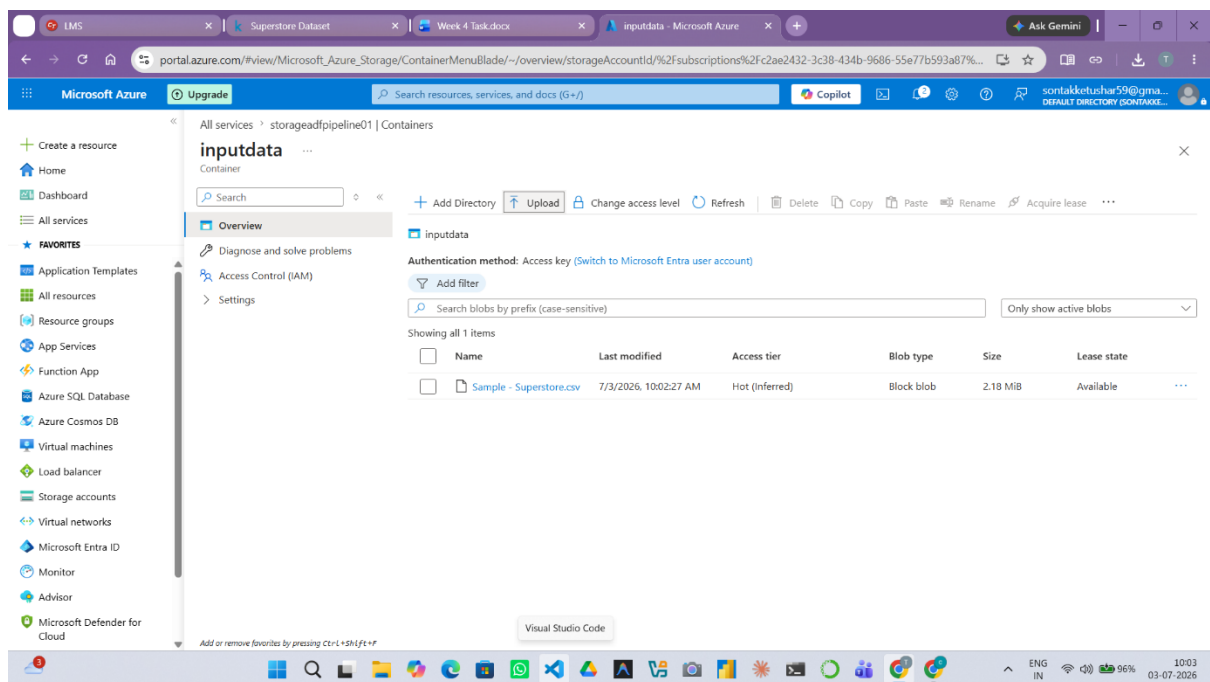
- Created a Storage Account.
- Created a Blob Container.
- Uploaded a CSV file.

Deliverable:

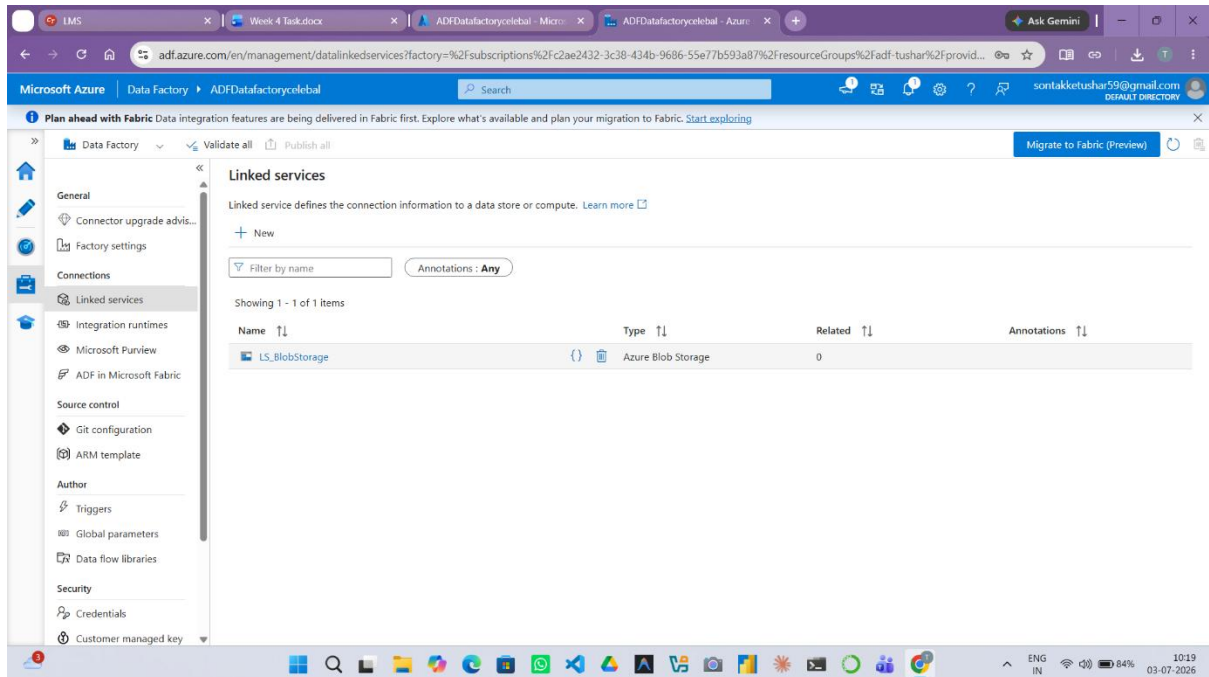
Description: Created a Storage Account, Blob Container, and uploaded the source CSV file.

Screenshot:

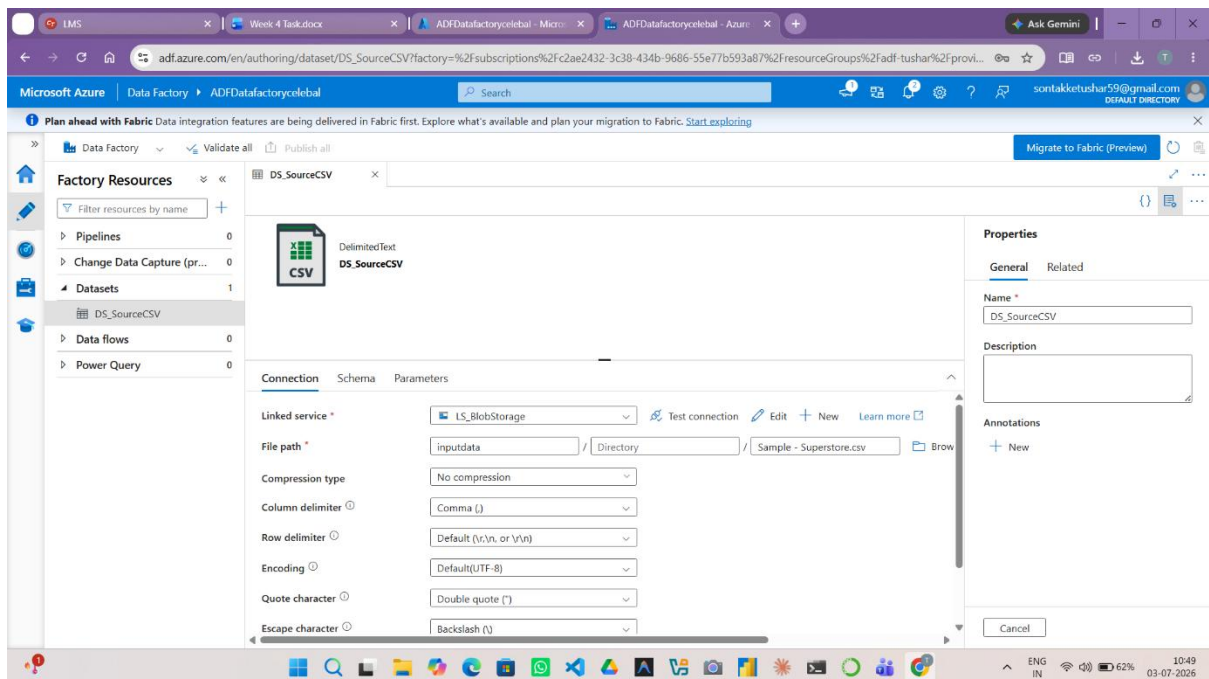
Created a Storage Account



- Created a Blob Container.



- Uploaded a CSV file.



Task 3: ADF Basics

Do:

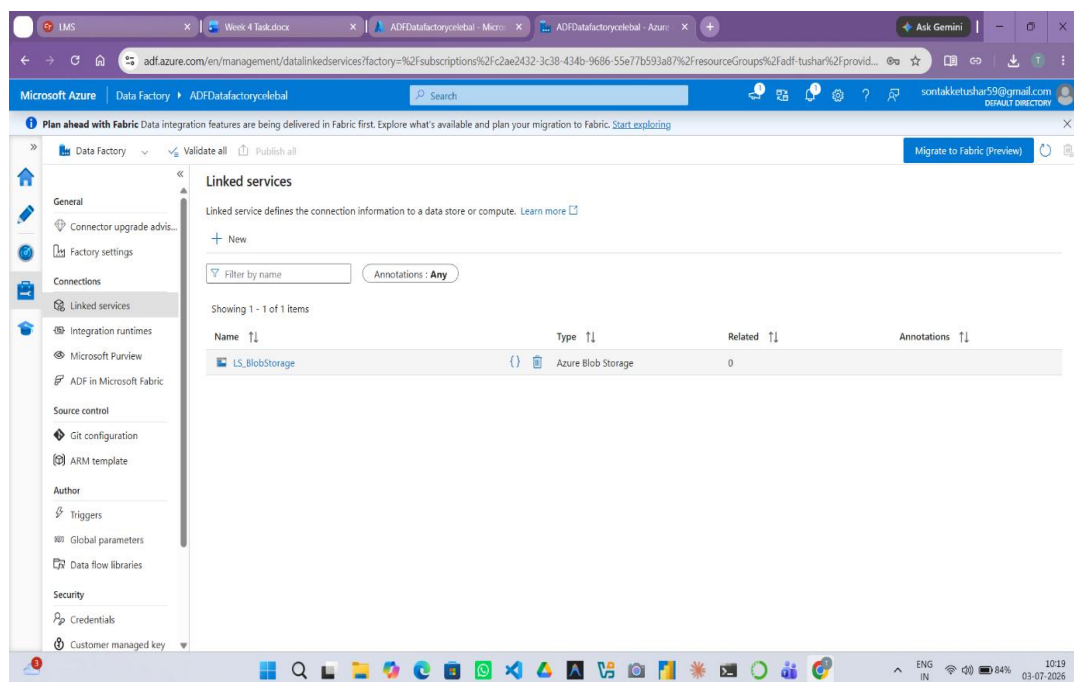
- Created Azure Data Factory.
- Explored ADF UI (Author, Monitor, Manage).
- Created Linked Service (Blob Storage).
- Created Source and Destination Datasets.
- Used Get Metadata activity.

Deliverable:

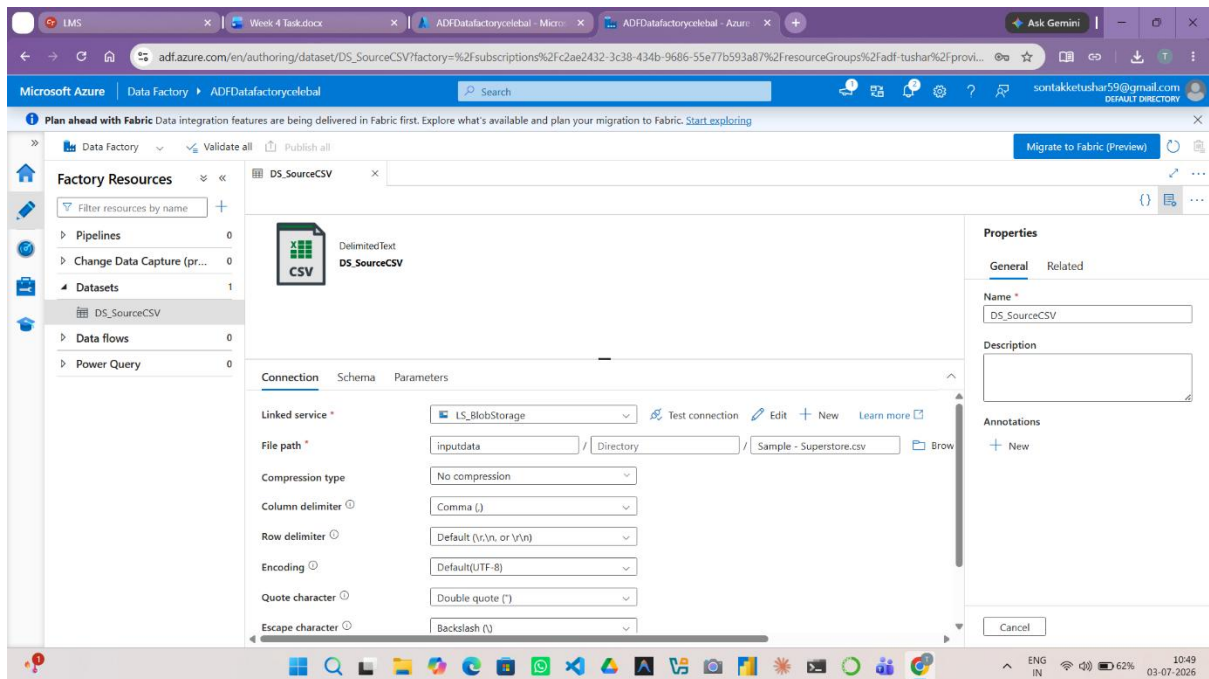
Description: Configured Azure Data Factory with Linked Service, Datasets, and Get Metadata activity.

Screenshots:

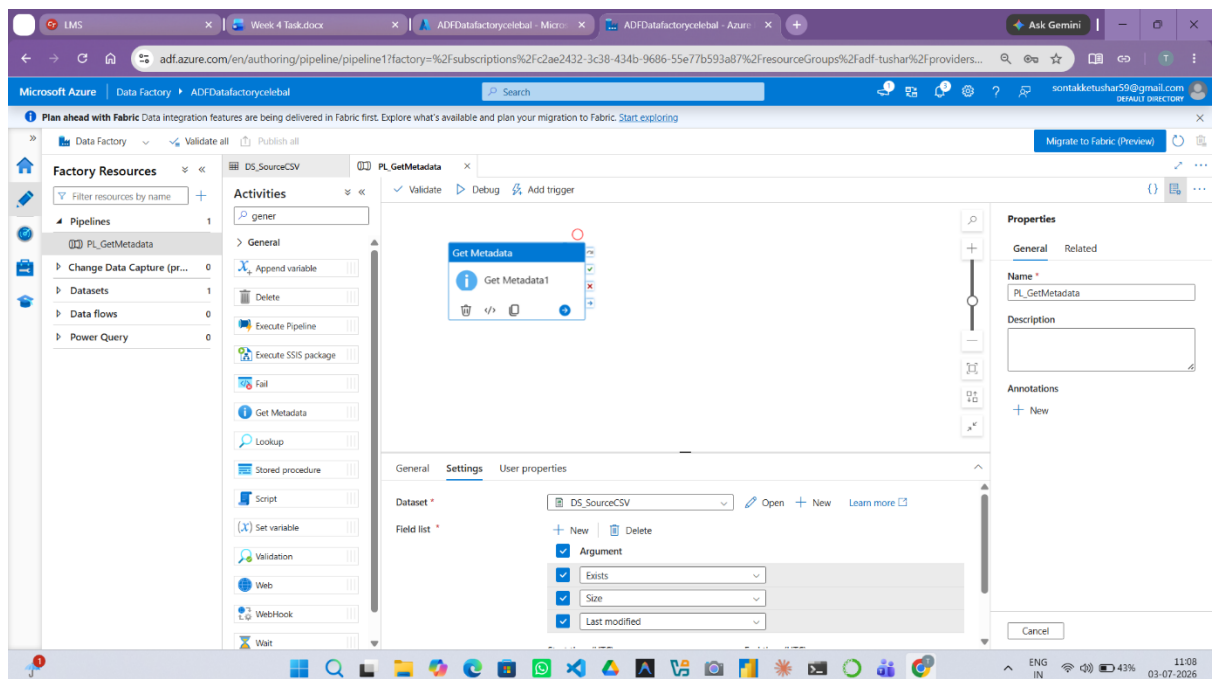
- Linked Service



- Dataset



- Get Metadata Activity



Task 4: Pipeline Development

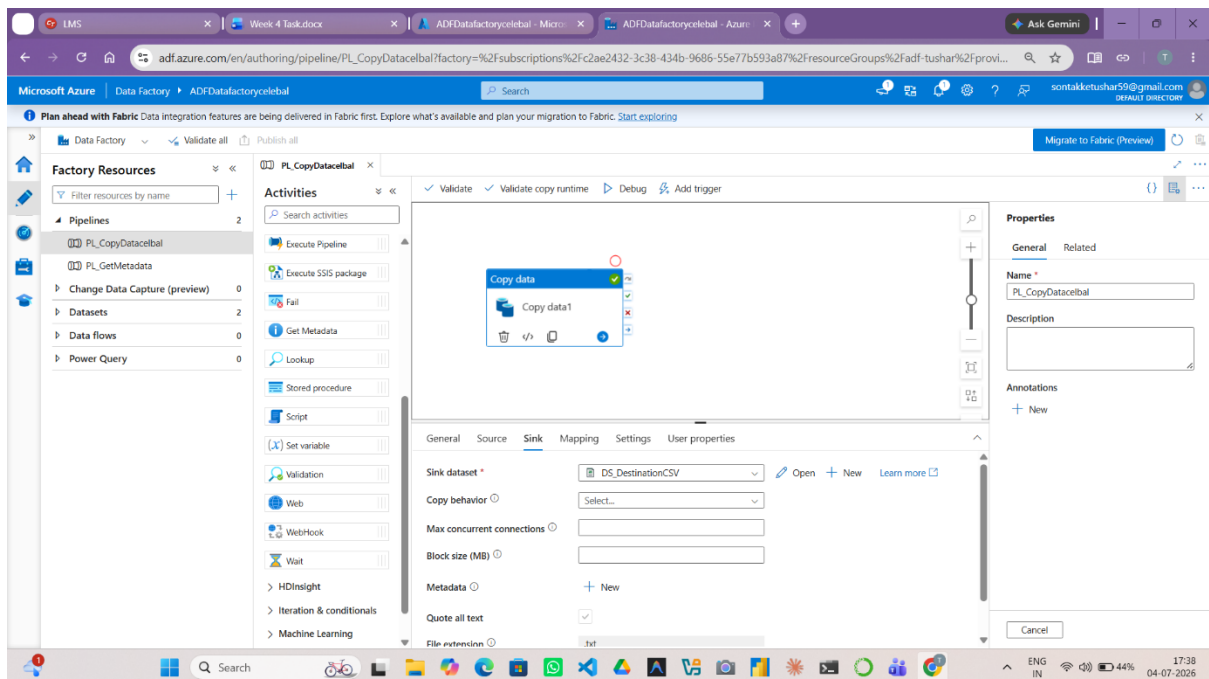
Do:

- Created a pipeline using Copy Data activity.
- Configured Source and Destination.

Deliverable:

Description: Developed a Copy Data pipeline with source and destination configuration.

Screenshot:



Task 5: Pipeline Execution

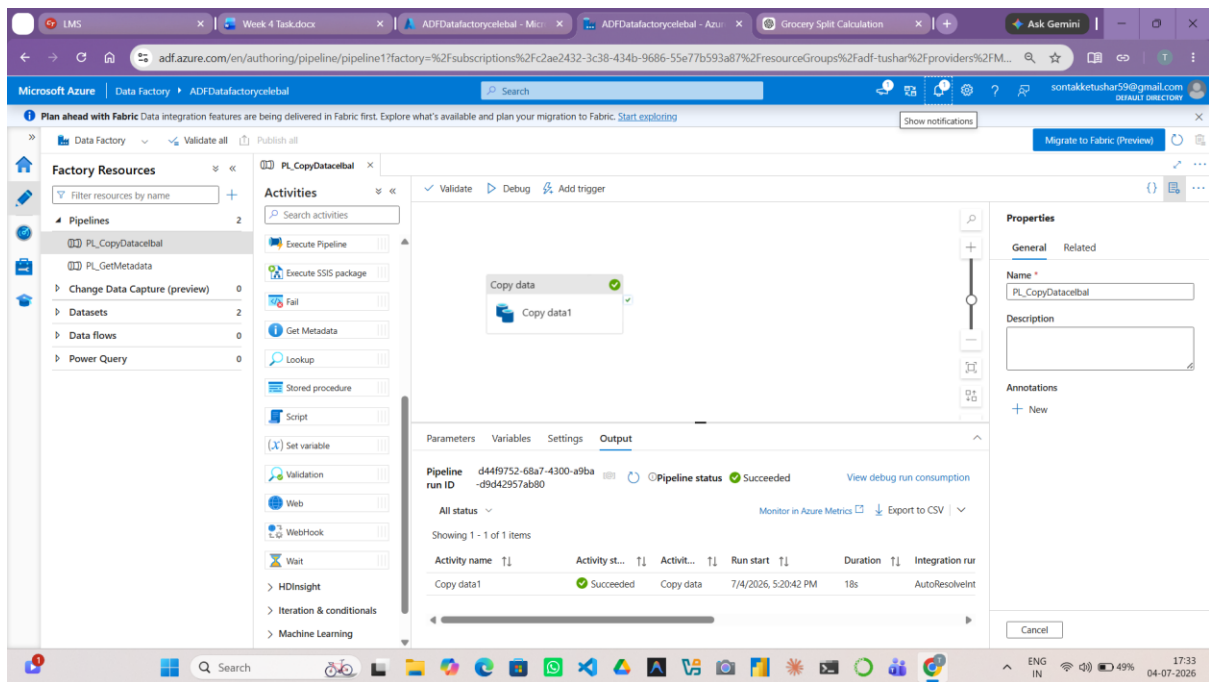
Do:

- Executed the pipeline using Debug/Trigger.

Deliverable:

Description: Successfully executed the pipeline and verified the successful status.

Screenshot:



Task 6: IAM Roles

Do:

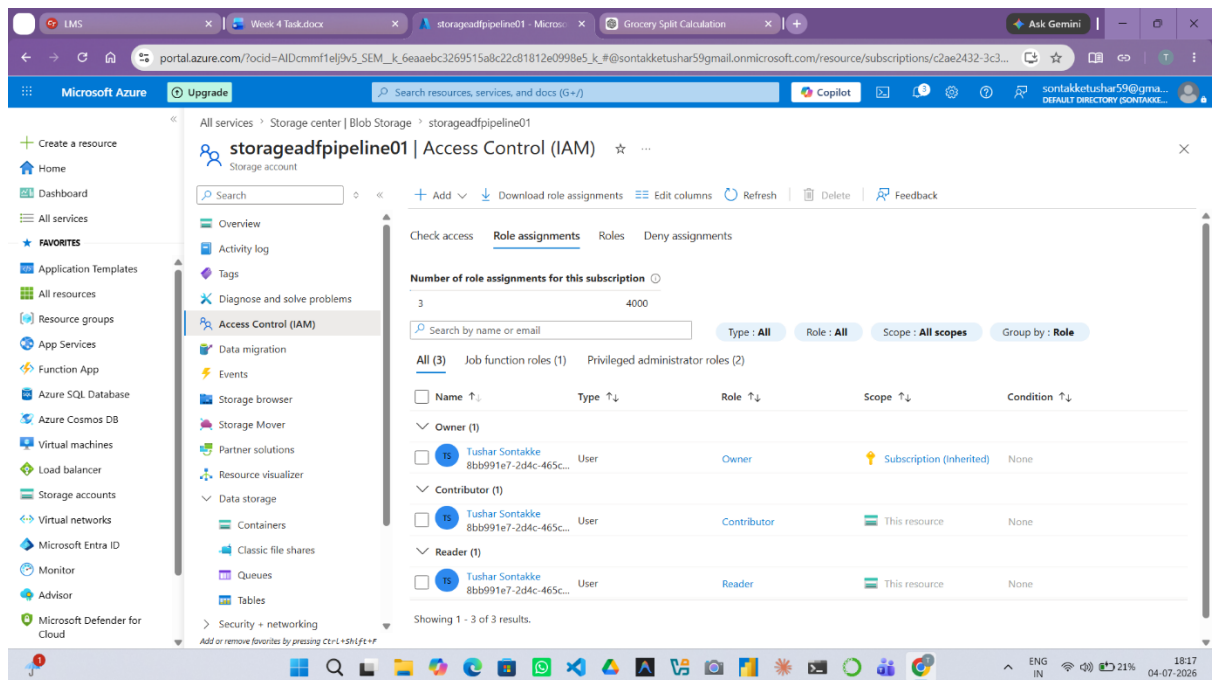
- Assigned Reader role.
- Assigned Contributor role.
- Provided Azure Data Factory access to Storage.

Deliverable:

Description: Assigned Reader and Contributor roles for Storage Account access.

Screenshots:

- Reader Role Assignment
- Contributor Role Assignment



Mini Project (Final Task)

Problem Statement

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements

- Source: CSV file in Blob Storage.
- Linked Service + Dataset + Pipeline.
- Process: Get Metadata + Copy Data.
- Destination: New CSV file in Blob Storage.

Implementation

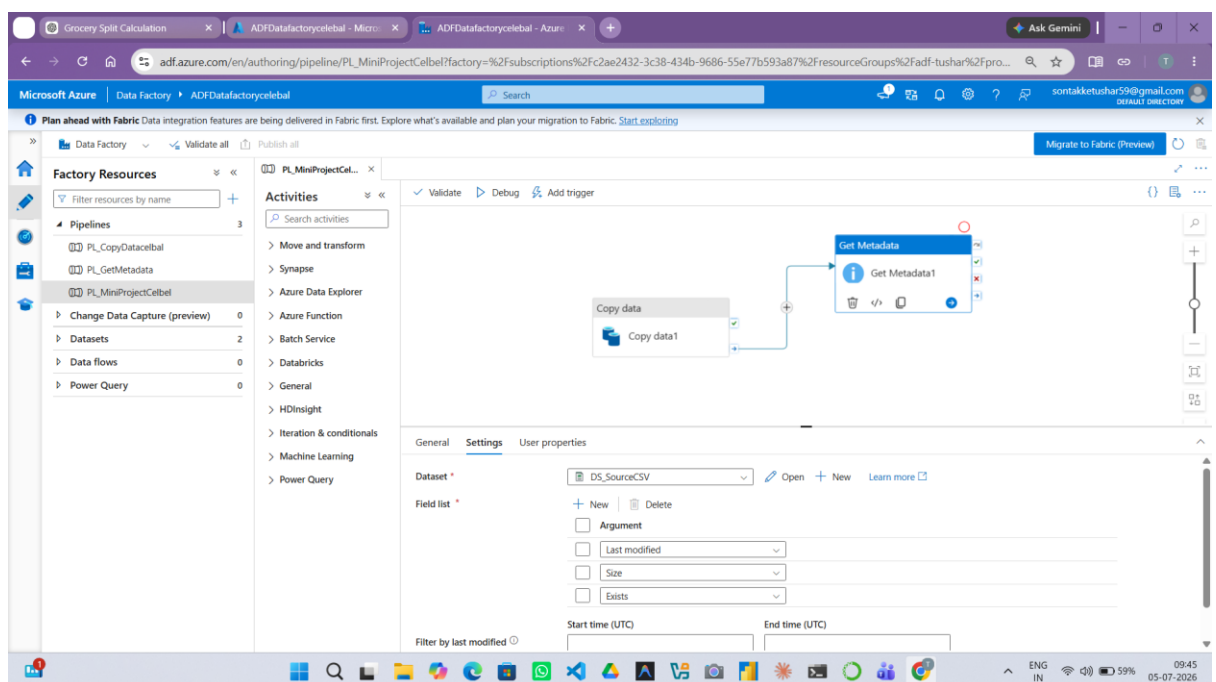
Description: Built an end-to-end Azure Data Factory pipeline that validated metadata using Get Metadata and copied the CSV file from the source to the destination Blob Storage.

Expected Output

- Pipeline executed successfully.
- Data copied to destination.
- Metadata validated successfully.

Screenshots

1. Mini Project Pipeline Design



2. Pipeline Execution (Succeeded)

The screenshot displays the Microsoft Azure Data Factory portal. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'PL_MiniProjectCelbel', 'PL_CopyDataCelbel', and 'PL_GetMetadata'. The main canvas shows a pipeline diagram with two activities: 'Copy data' and 'Get Metadata'. The 'Copy data' activity is connected to 'Copy data1', which is connected to 'Get Metadata1'. The 'Get Metadata1' activity is connected to 'Get Metadata1'. The 'Pipeline status' is 'Succeeded'. The 'Output' tab is selected, showing a table of activity results.

Activity name	Activity st...	Activit...	Run start	Duration	Integration ru...
Get Metadata1	Succeeded	Get Metadata	7/4/2026, 6:34:26 PM	16s	AutoResolveInt
Copy data1	Succeeded	Copy data	7/4/2026, 6:34:03 PM	22s	AutoResolveInt

3. Output.csv in Destination

The screenshot displays the Microsoft Azure Storage Explorer. The left sidebar shows the 'All services' tree with 'Storage accounts' expanded, listing 'storageadfpipeline01'. The main canvas shows the 'inputdata' container. The 'Overview' tab is selected, showing a table of blobs. The 'Output.csv' blob is highlighted.

Name	Last modified	Access tier	Blob type	Size	Lease state
Output.csv	7/4/2026, 6:35:38 PM	Hot (Inferred)	Block blob	2.18 MiB	Available
Sample - Superstore.csv	7/3/2026, 10:02:27 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

Microsoft Azure portal interface showing the 'Output.csv' blob details.

Left Sidebar (Navigation):

- Create a resource
- Home
- Dashboard
- All services
- FAVORITES
- Application Templates
- All resources
- Resource groups
- App Services
- Function App
- Azure SQL Database
- Azure Cosmos DB
- Virtual machines
- Load balancer
- Storage accounts
- Virtual networks
- Microsoft Entra ID
- Monitor
- Advisor
- Microsoft Defender for Cloud

Top Bar: Microsoft Azure | Upgrade | Search resources, services, and docs (G + J) | Copilot | User profile: sontakkotushar59@gmail.com

Main Content Area:

inputdata Container

Output.csv Blob

Actions: Save, Discard, Download, Refresh, Delete, Change tier, Acquire lease, Break lease, Give feedback

Overview | Versions | Snapshots | Edit | Generate SAS

Properties:

URL	https://storageadfpipeli...
LAST MODIFIED	4/7/2026, 6:35:38 pm
CREATION TIME	4/7/2026, 5:20:21 pm
VERSION ID	-
TYPE	Block blob
SIZE	2.18 MiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DED9CCF18BCB62
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/octet-stream
CONTENT-MD5	swZpBenrTud0U9Rz2/IMAg...
CONTENT-ENCODING	
CONTENT-LANGUAGE	

Bottom Taskbar: Windows taskbar with icons for Start, Search, File Explorer, Microsoft Store, Edge, Teams, OneDrive, and system tray showing ENG IN, 69% battery, and date 04-07-2026.